

Faculty Of Engineering and Technology

Electrical and Computer Engineering Department

ENEE 4113, Communication Laboratory

Pulse Amplitude Modulation (PAM)

Pre-Lab Experiment #6

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Section: 5.

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Part 1: Time and Frequency Characteristics of pulse train:

Pulse Train:

Block Diagram:

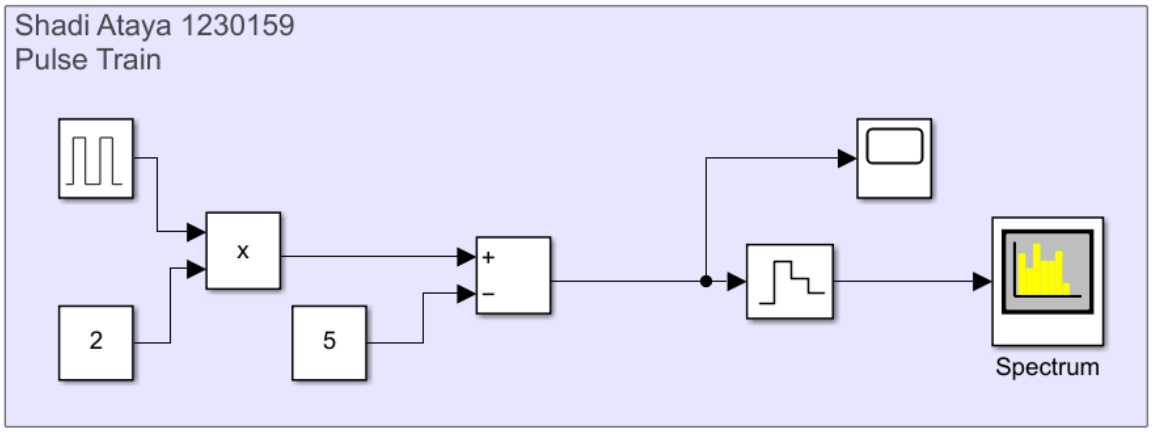


Figure 1: Pulse Train Block Diagram

Pulse Train at duty cycle 10% in time and frequency domain

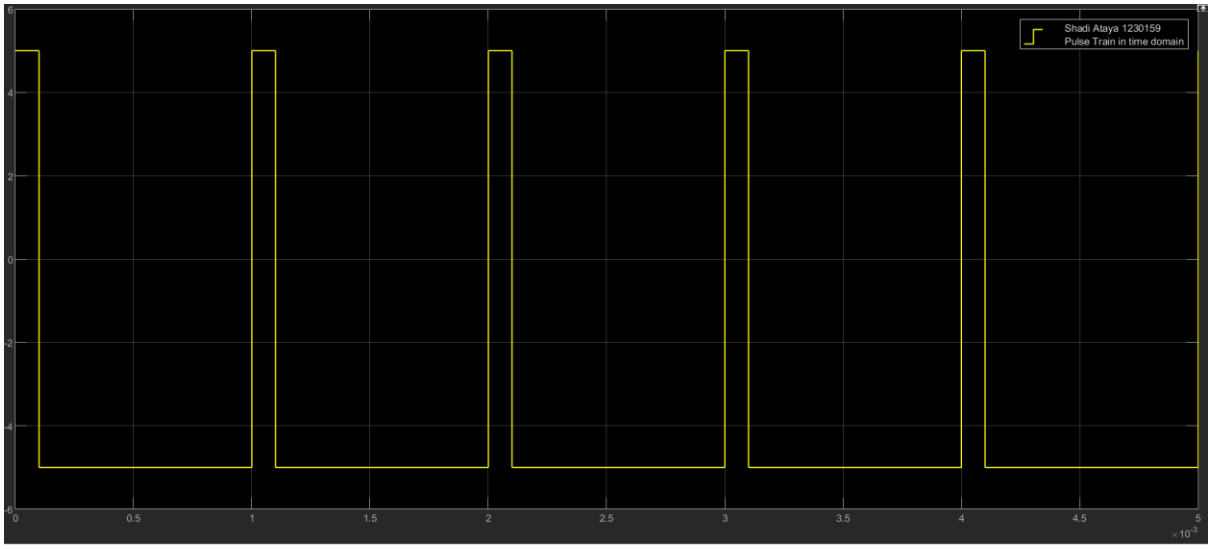


*Figure 2: Pulse Train at duty cycle **10%** in time domain*

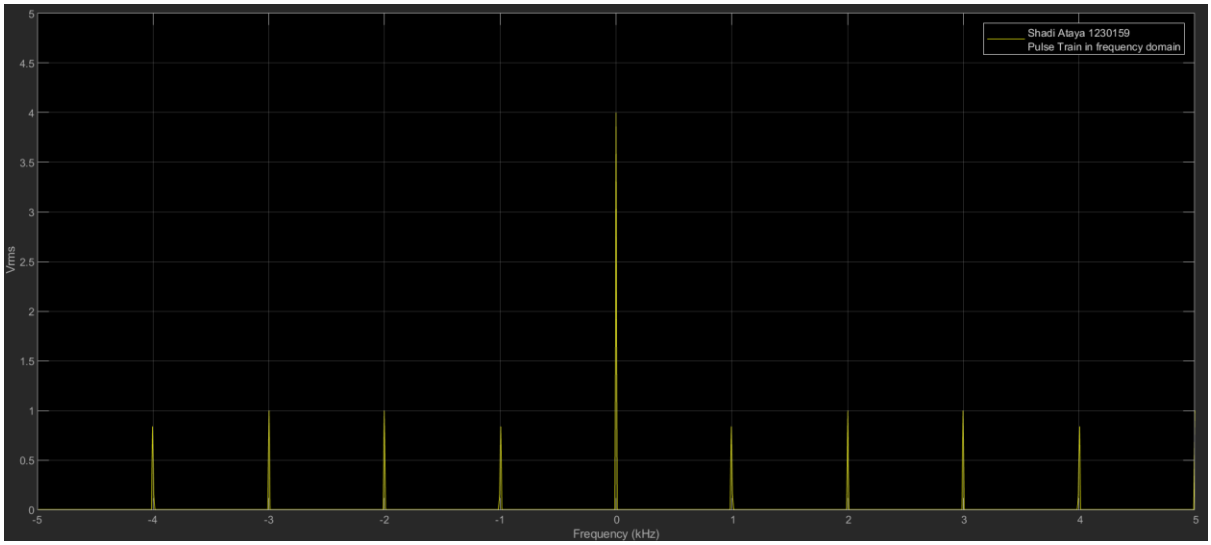


*Figure 3: Pulse Train at duty cycle **10%** in frequency domain*

Pulse Train at duty cycle 50% in time and frequency domain

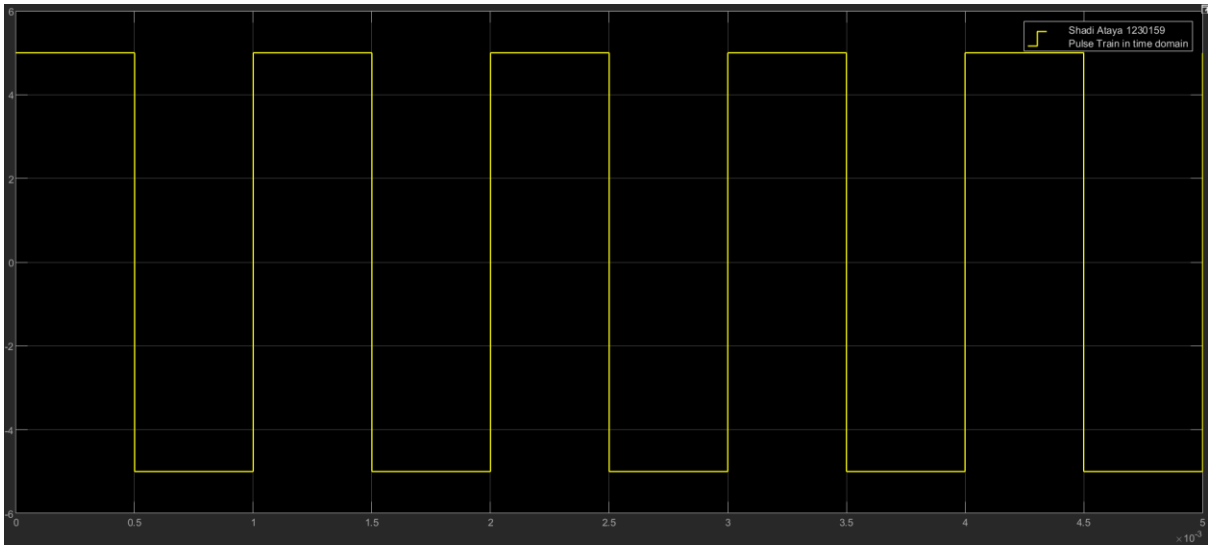


*Figure 4: Pulse Train at duty cycle **50%** in time domain*

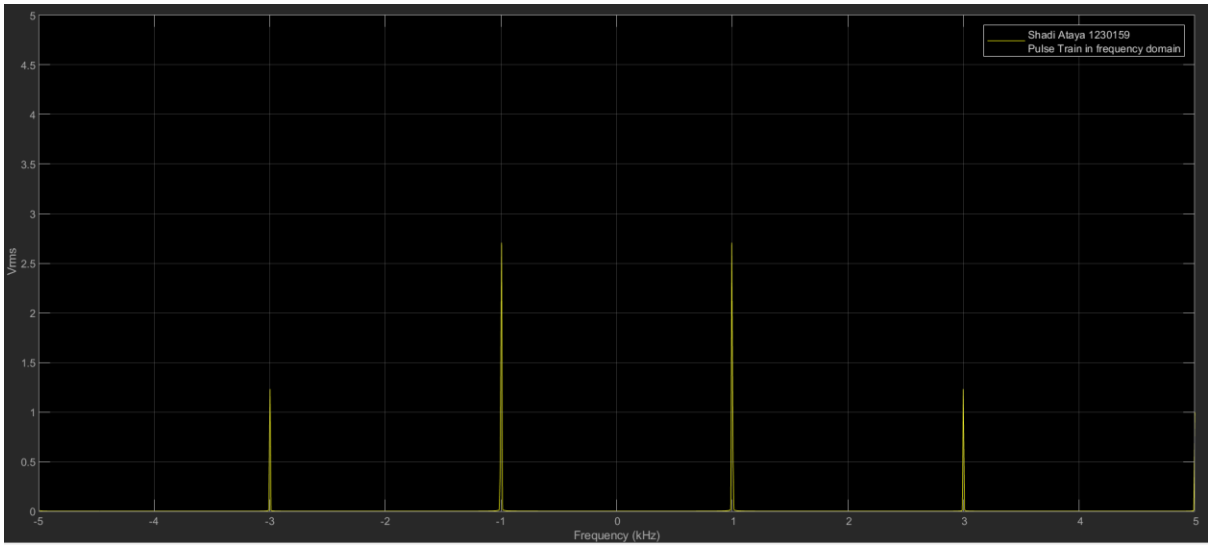


*Figure 5: Pulse Train at duty cycle **50%** in frequency domain*

Part 2: Natural Sampling Modulation and demodulation

Message Signal:

Block Diagram:

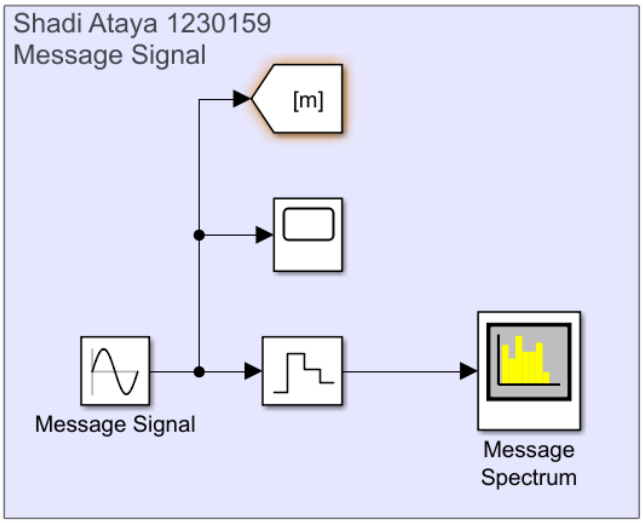


Figure 6: Message Signal Block Diagram

Message Signal in time and frequency domain:

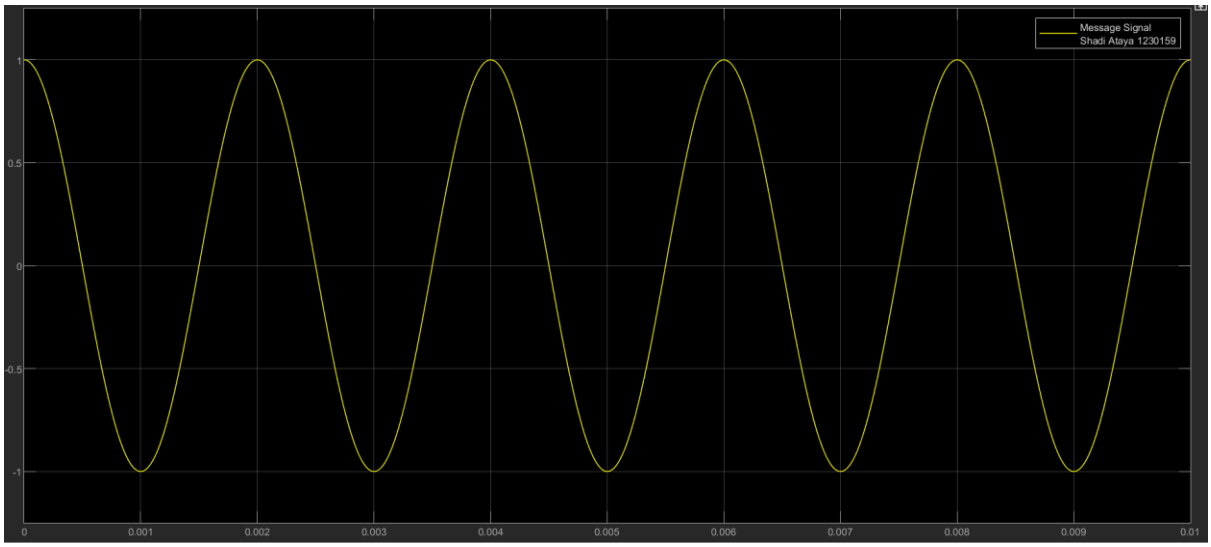


Figure 7: Message Signal frequency = 500 in time domain

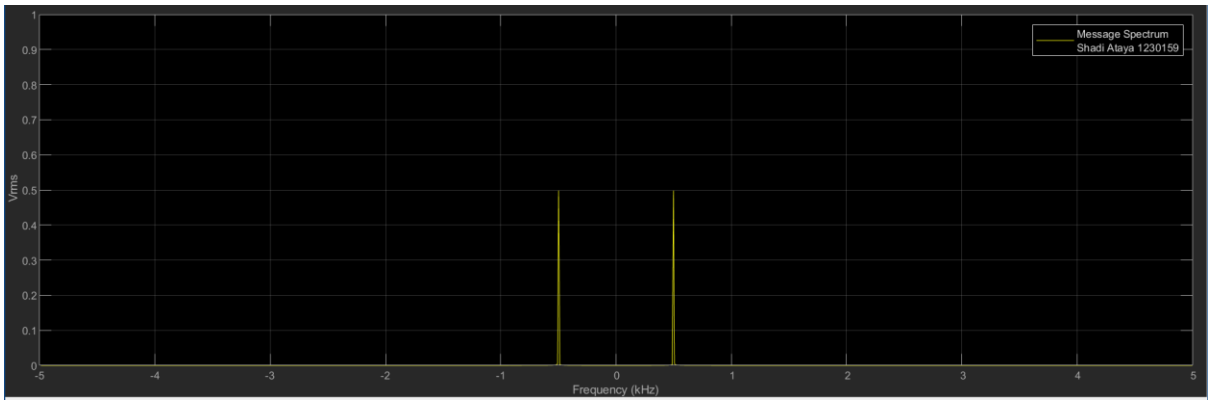


Figure 8: Message Signal frequency = 500 in frequency domain

Natural Sampling modulation:

Block Diagram:

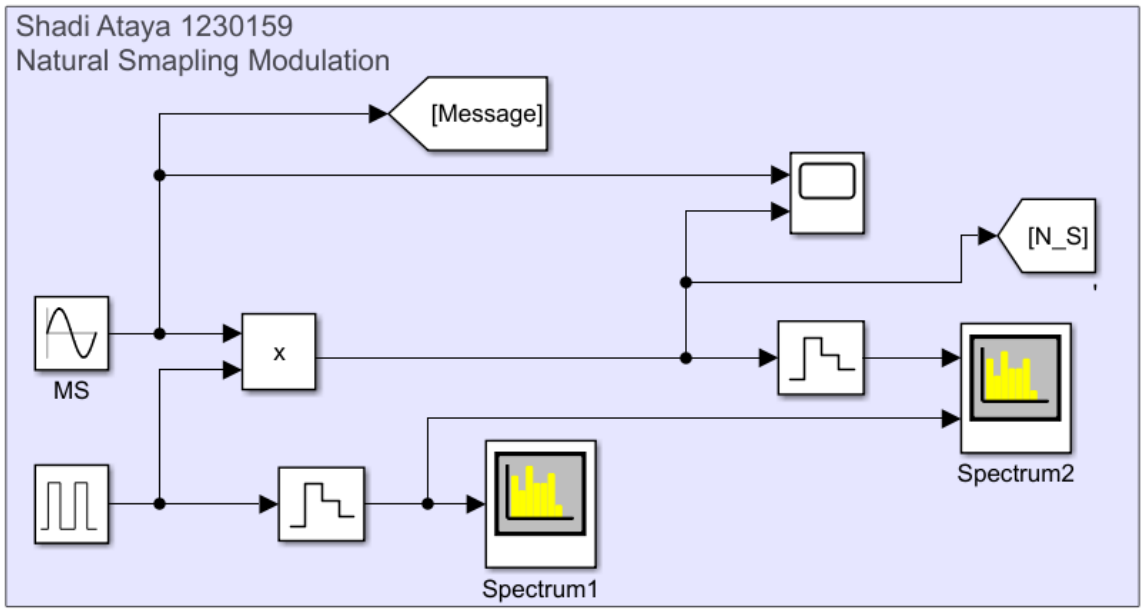


Figure 9: Natural Sampling modulation

Modulated signal at frequency = 500Hz & D= 50% in time and frequency domain

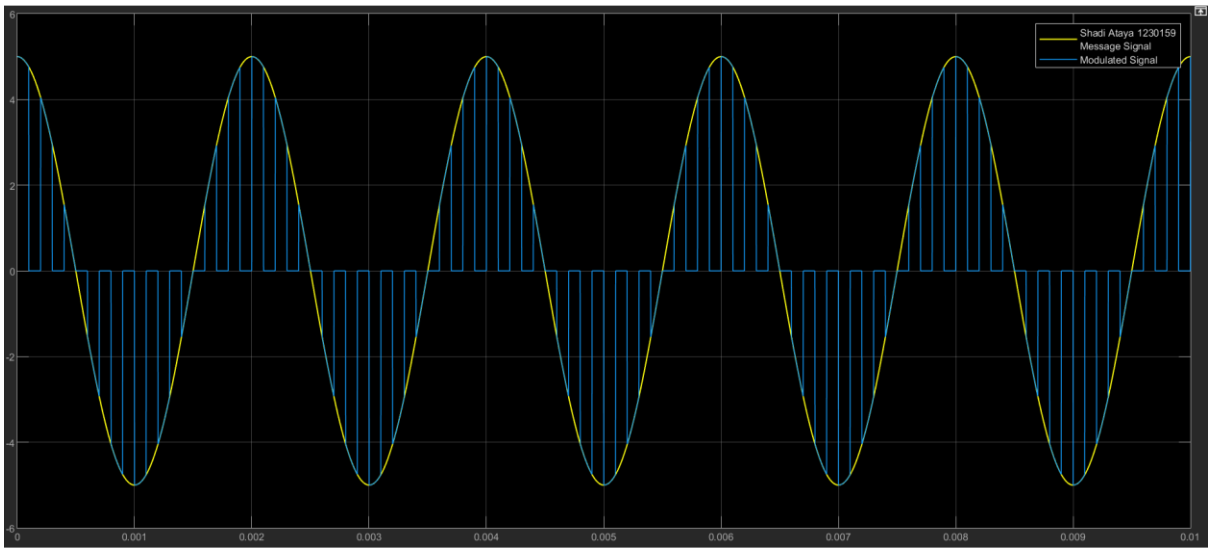


*Figure 10: Modulated signal at frequency =**500** and D = **50%** in time domain*

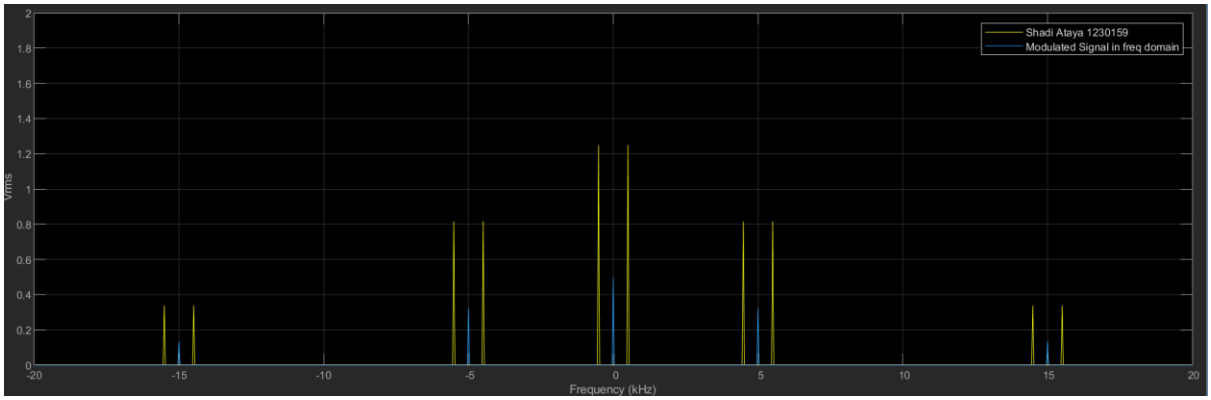


*Figure 11: Modulated signal at frequency =**500** and D = **50%** in frequency domain*

Modulated signal at frequency = 1000 and D = 50% in time domain and frequency domain

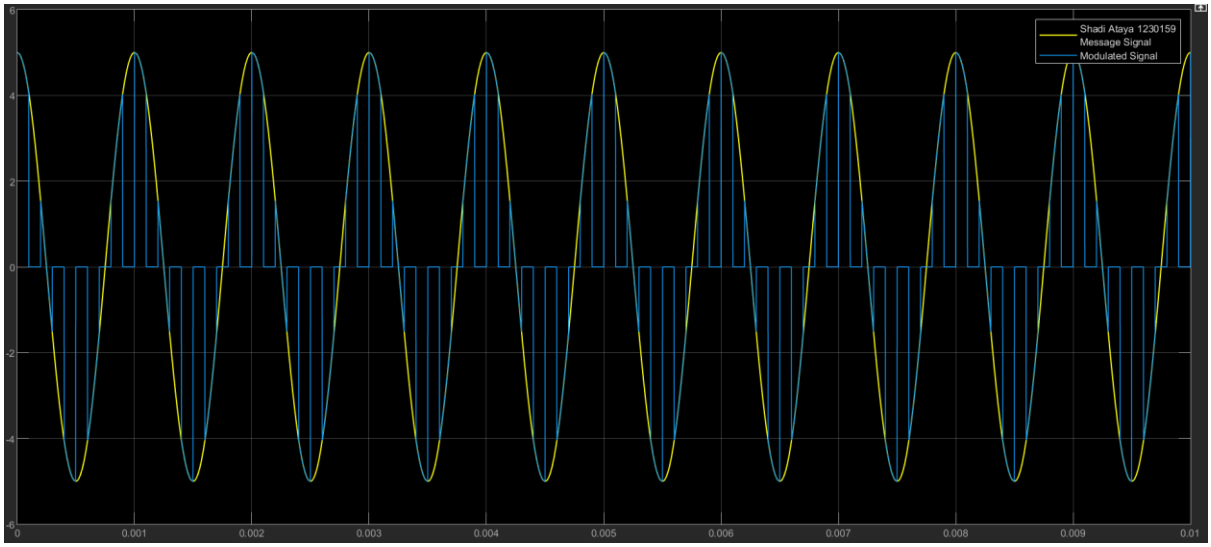


*Figure 12: Modulated signal at frequency =**1000** and D = **50%** in time domain*

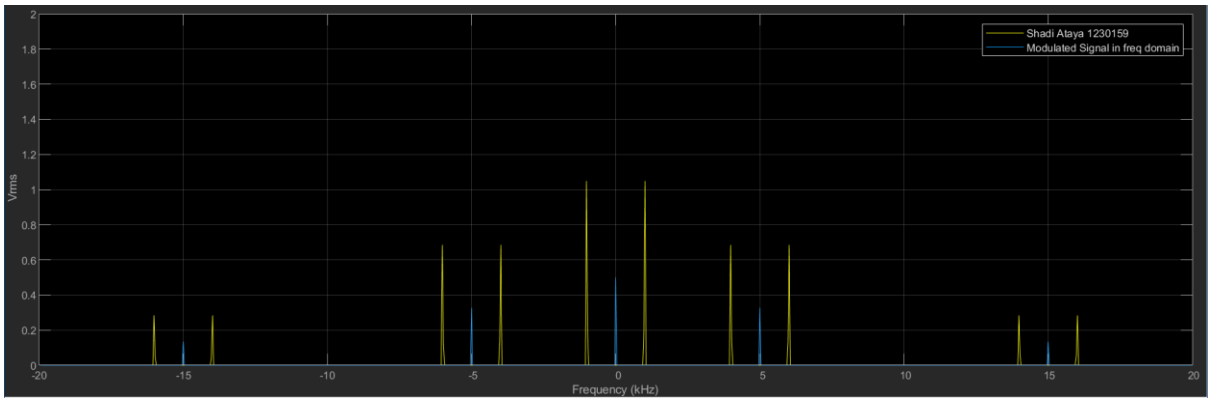


*Figure 13: Modulated signal at frequency =**1000** and D = **50%** in frequency domain*

Modulated signal at frequency = 2000 and D = 50% in time domain and frequency domain

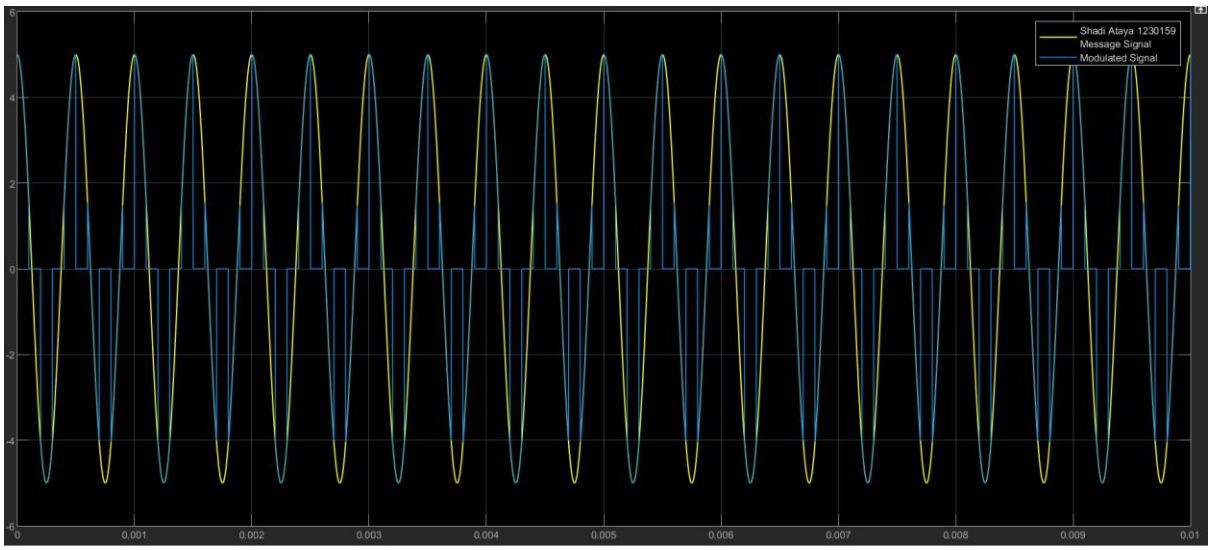


*Figure 14: Modulated signal at frequency =**2000** and D = **50%** in time domain*

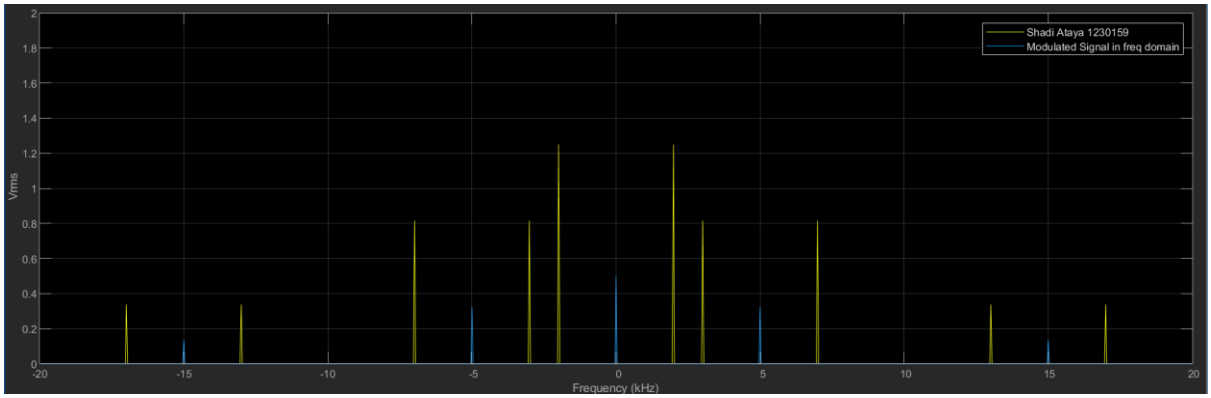


*Figure 15: Modulated signal at frequency =**2000** and D = **50%** in frequency domain*

Modulated signal at frequency = 500 and D = 10% in time domain and frequency domain

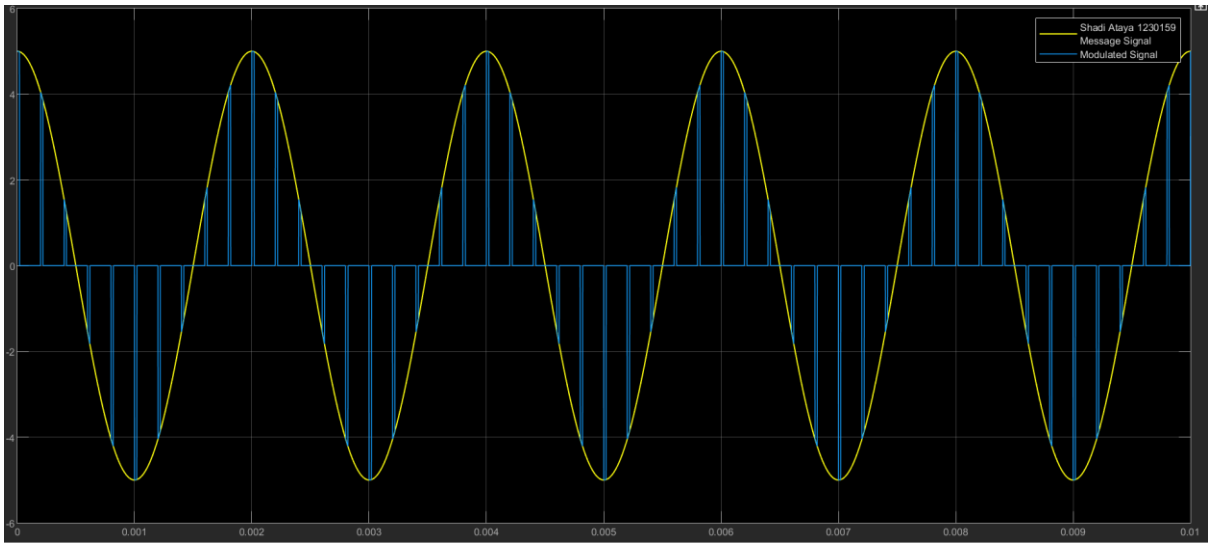


*Figure 16: Modulated signal at frequency =**500** and D = **10%** in time domain*

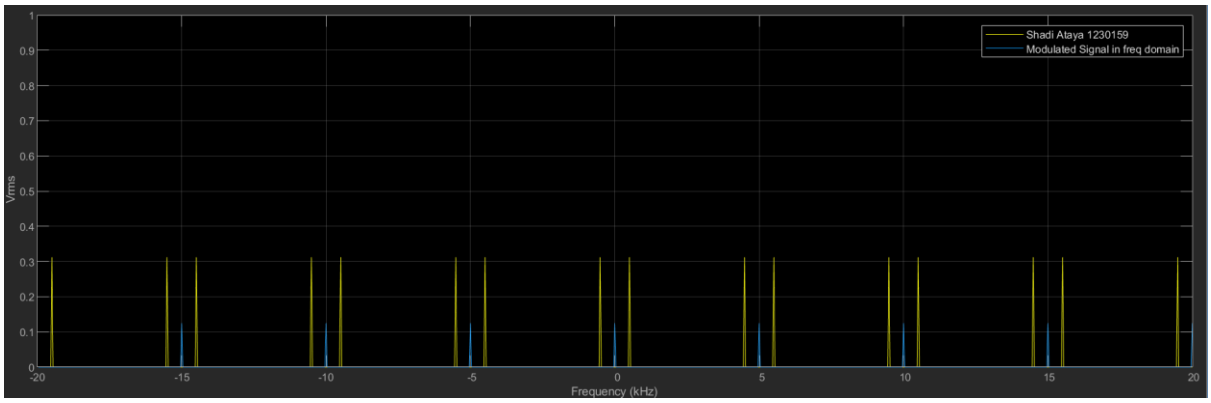


*Figure 17: Modulated signal at frequency =**500** and D **= 10%** in frequency domain*

Modulated signal at frequency = 500 and D = 30% in time domain and frequency domain

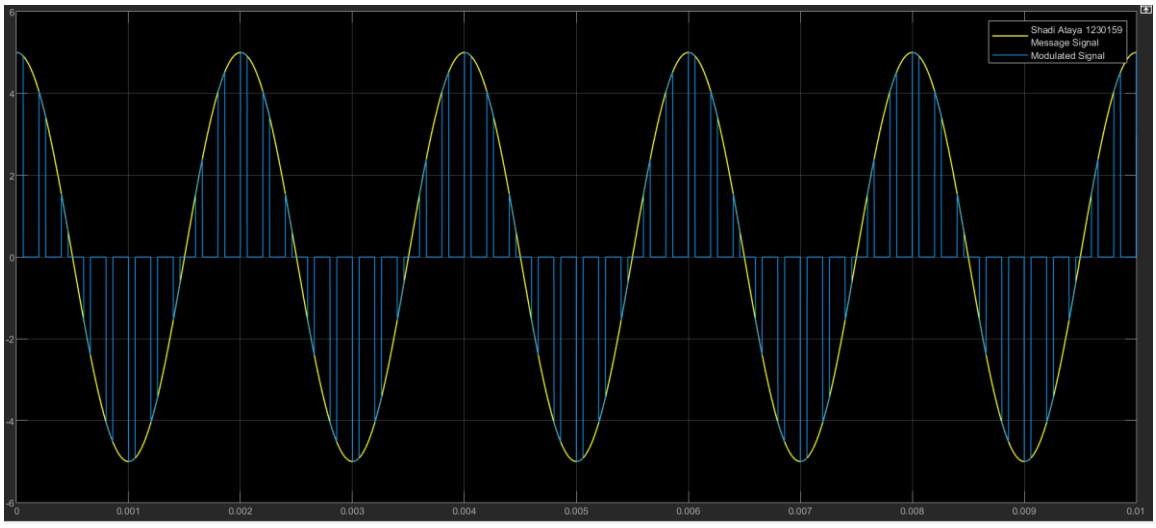


*Figure 18: Modulated signal at frequency =**500** and D = **30%** in time domain*

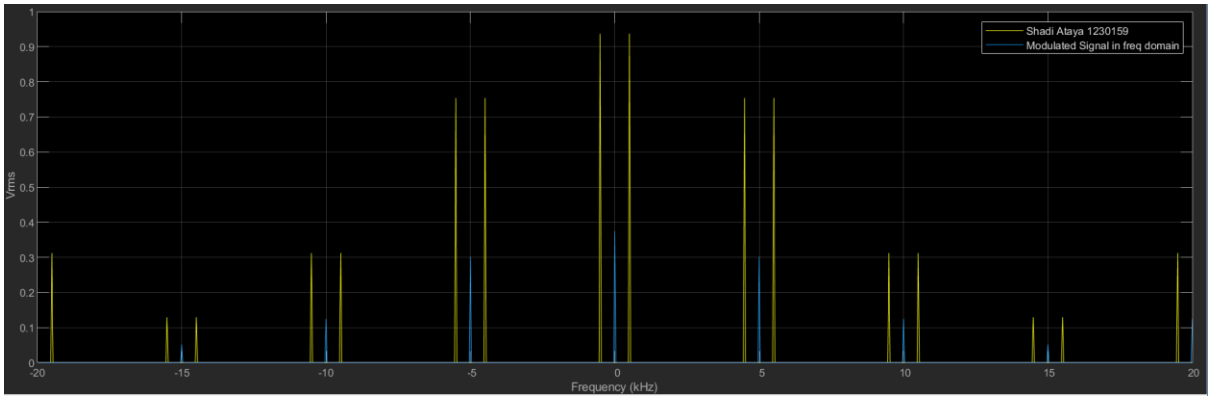


*Figure 19: Modulated signal at frequency =**500** and D = **30%** in frequency domain*

From Part 4: Modulated signal at frequency = 3000 and D = 50% in time domain and frequency domain

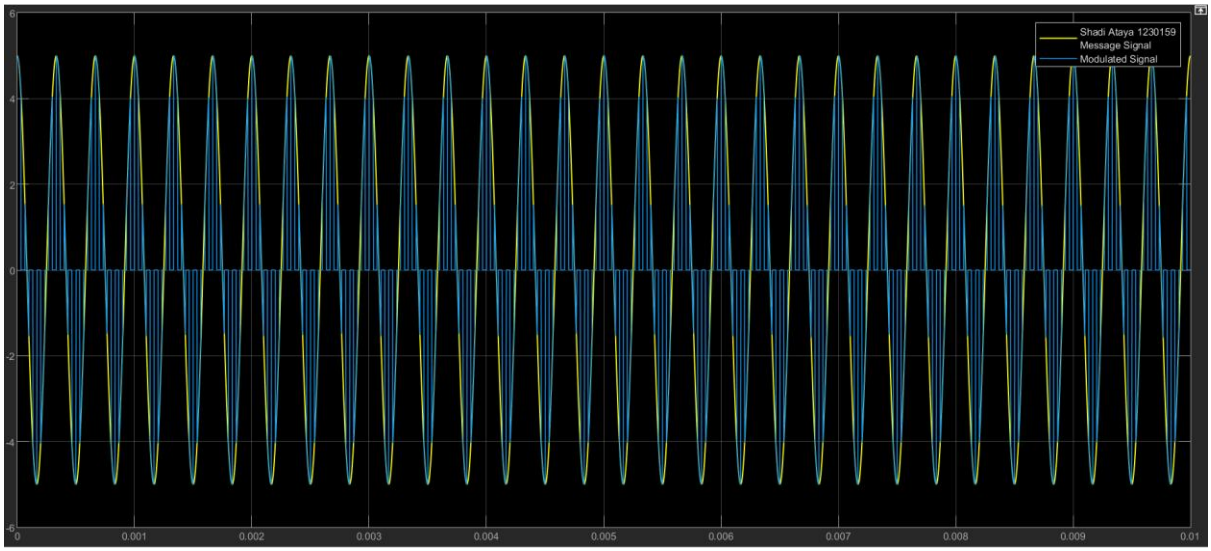


*Figure 20: Modulated signal at frequency = **3000** and D = **50%** in time domain*

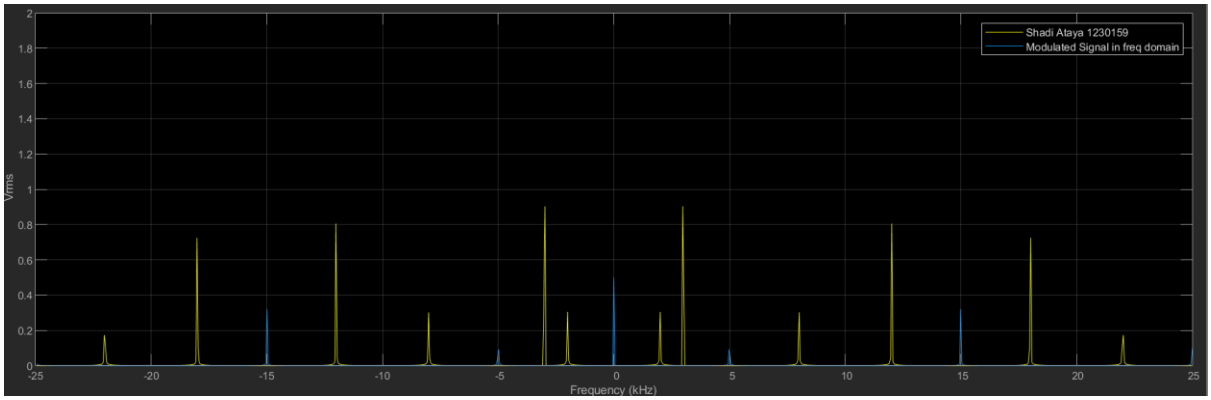


*Figure 21: Modulated signal at frequency = **3000** and D = **50%** in frequency domain*

Natural Sampling demodulation:

Block Diagram:

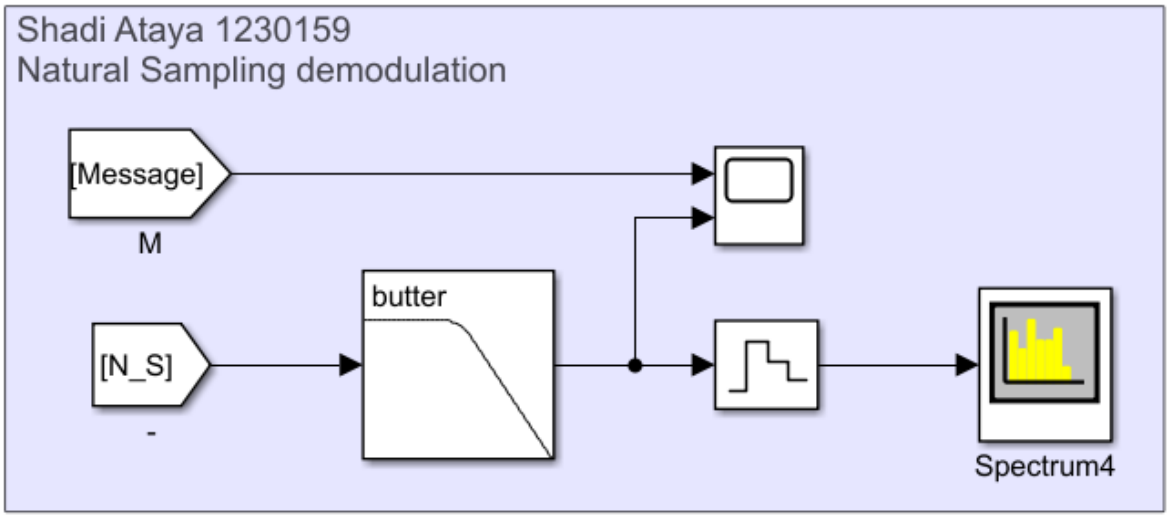


Figure 22: Natural sampling demodulation

Demodulated signal at frequency=500 and D =50% in time domain and frequency domain

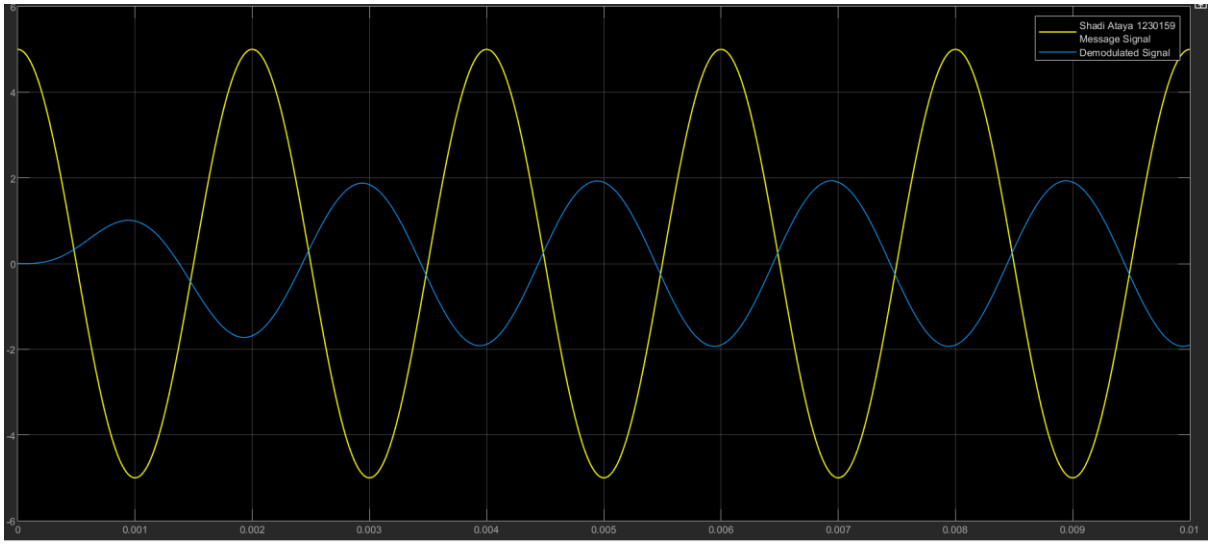


*Figure 23: Demodulated signal at frequency=**500** and D =**50%** in time domain*

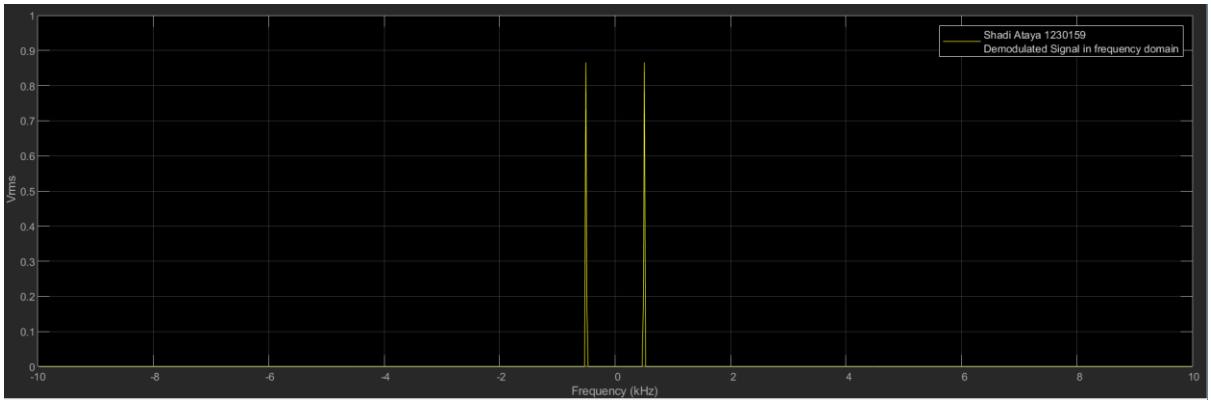


Figure 24: Demodulated signal at frequency=500 and D =50% in frequency domain

From Part 4: demodulated signal at frequency = 3000 and D = 50% in time domain and frequency domain

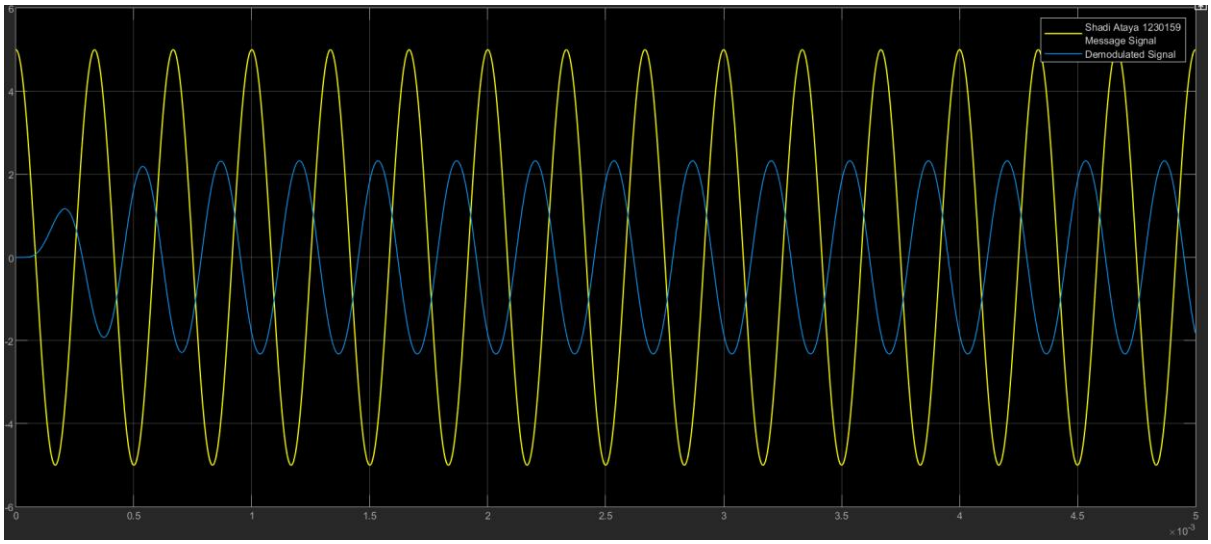


*Figure 25: Demodulated signal at frequency=**3000** and D =**50%** in time domain*

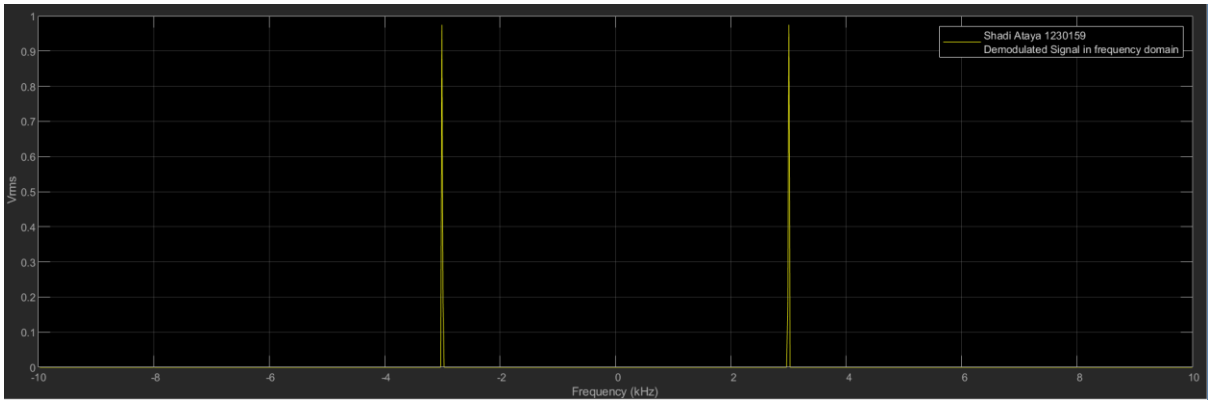


*Figure 26: Demodulated signal at frequency=**3000** and D =**50**% in frequency domain*

Part 3: Sample and Hold (Flat-top) Sampling Modulation and demodulation

Sample and Hold (Flat-top) Sampling Modulation

Block Diagram:

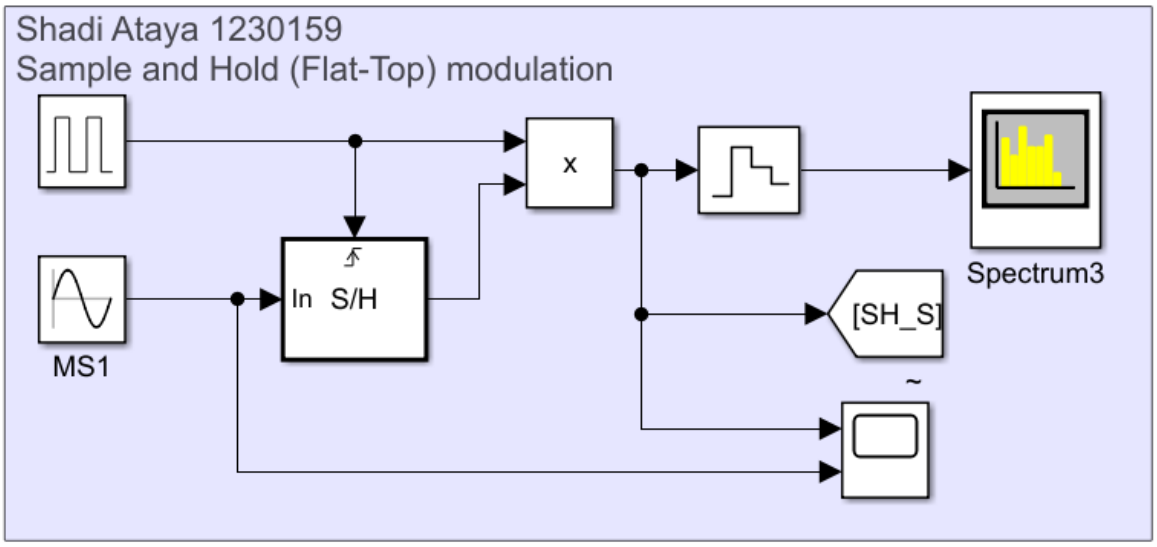


Figure 27: Sample and Hole (Flat-Top) modulation

Modulated signal at frequency = 500Hz & D= 50% in time and frequency domain

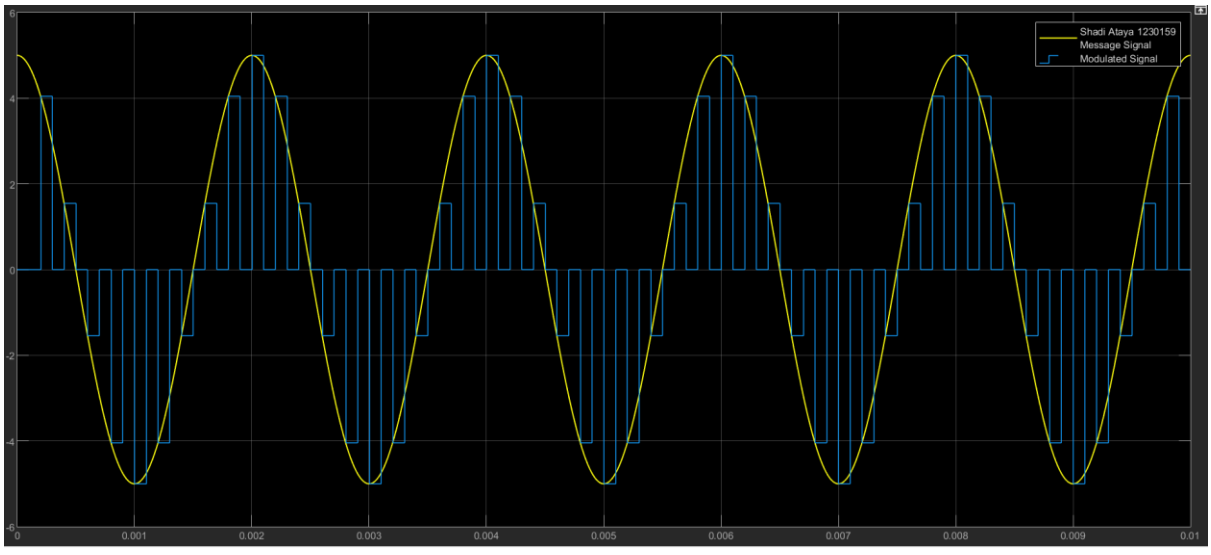


Figure 28: Modulated signal at frequency = 500Hz & D= 50% in time domain

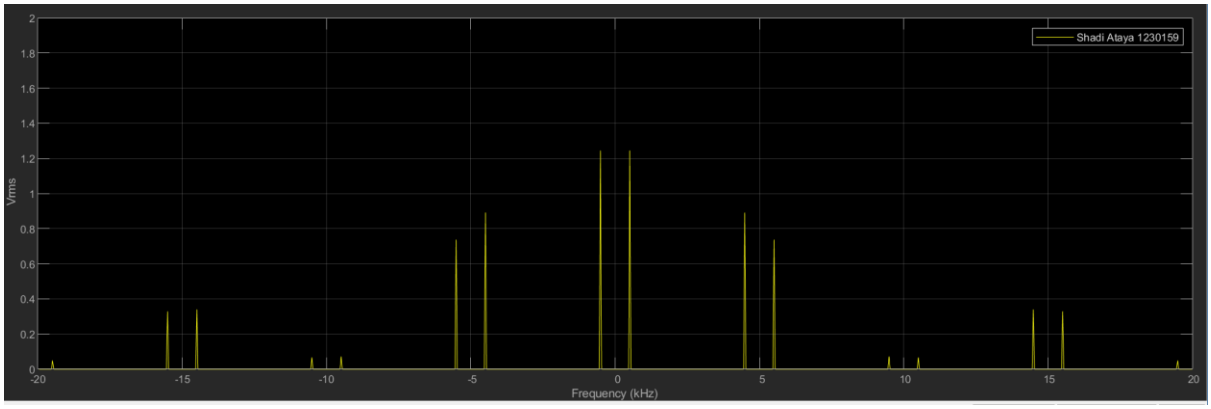


*Figure 29: Modulated signal at frequency = **500Hz** & D= **50%** in frequency domain*

Modulated signal at frequency = 1000 and D = 50% in time domain and frequency domain

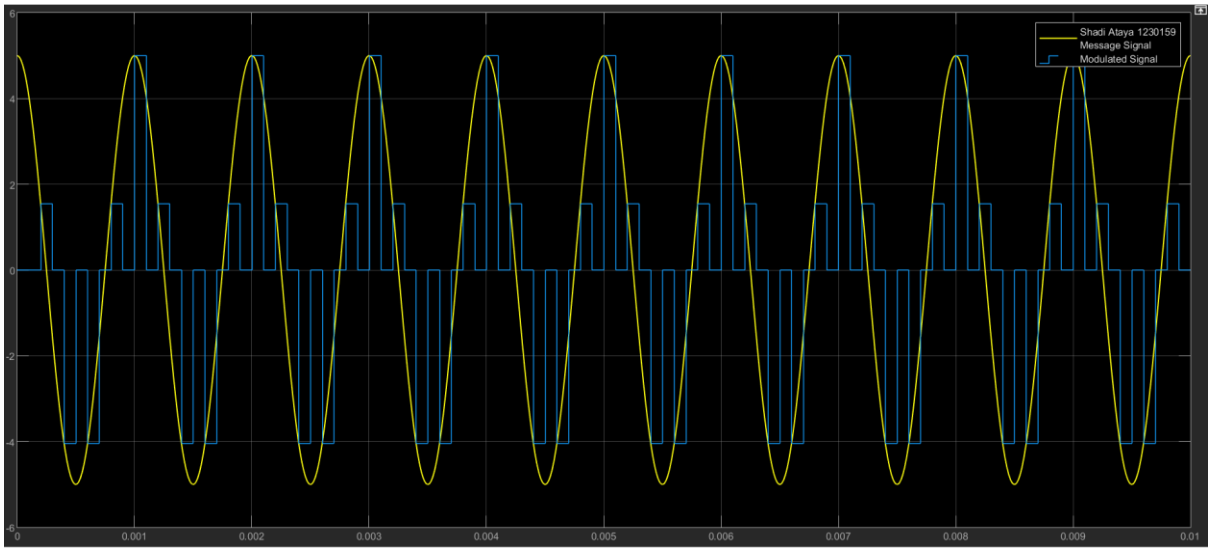


*Figure 30: Modulated signal at frequency = **1000Hz** & D= **50%** in time domain*

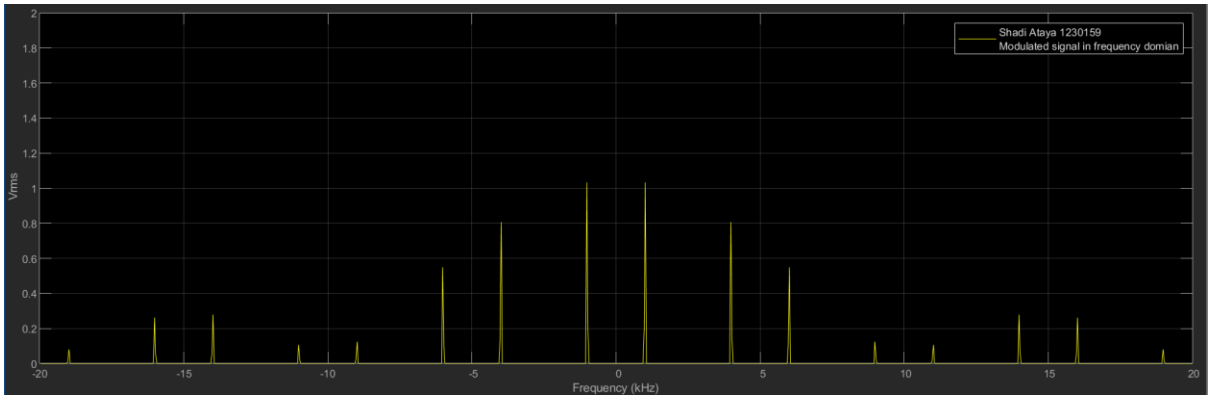


*Figure 31: Modulated signal at frequency = **1000Hz** & D= **50%** in frequency domain*

Modulated signal at frequency = 2000 and D = 50% in time domain and frequency domain

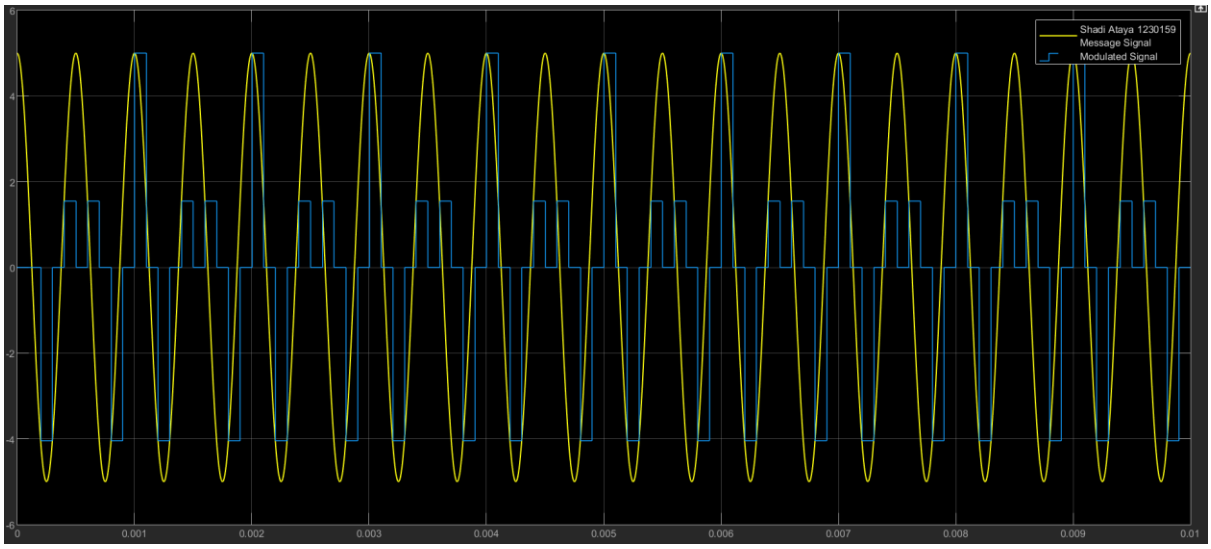


*Figure 32: Modulated signal at frequency = **2000Hz** & D= **50%** in time domain*

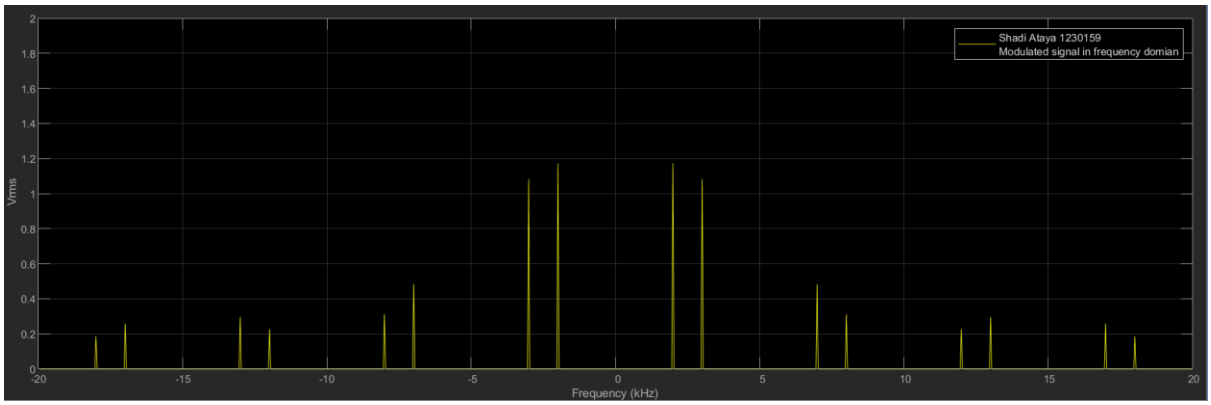


*Figure 33: Modulated signal at frequency = **2000Hz** & D= **50%** in frequency domain*

Modulated signal at frequency = 500 and D = 10% in time domain and frequency domain

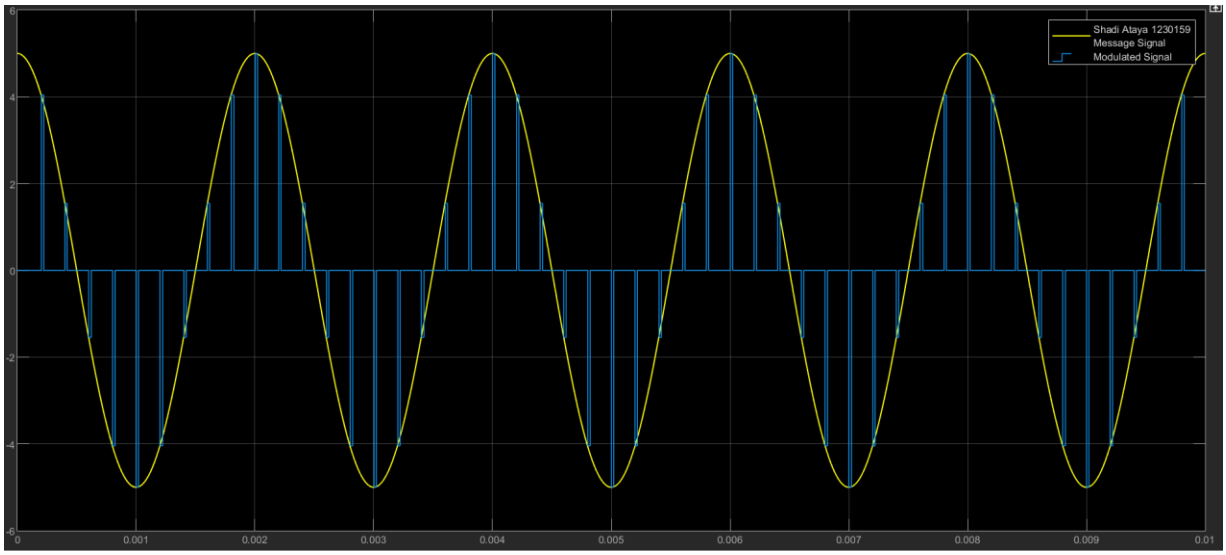


*Figure 34: Modulated Signal at frequency = **500Hz** and D = **10%** in time domain*

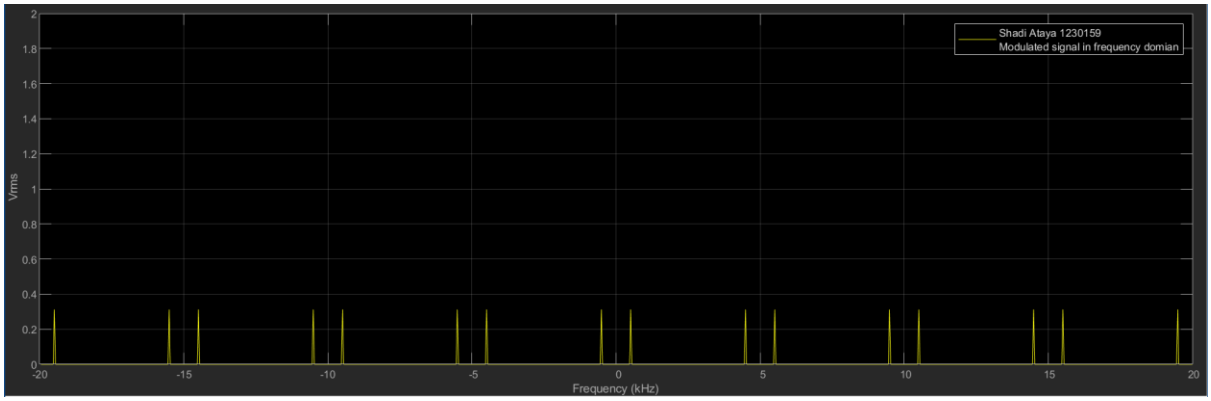


*Figure 35: Modulated Signal at frequency = **500Hz** and D = **10%** in frequency domain*

Modulated signal at frequency = 500 and D = 30% in time domain and frequency domain

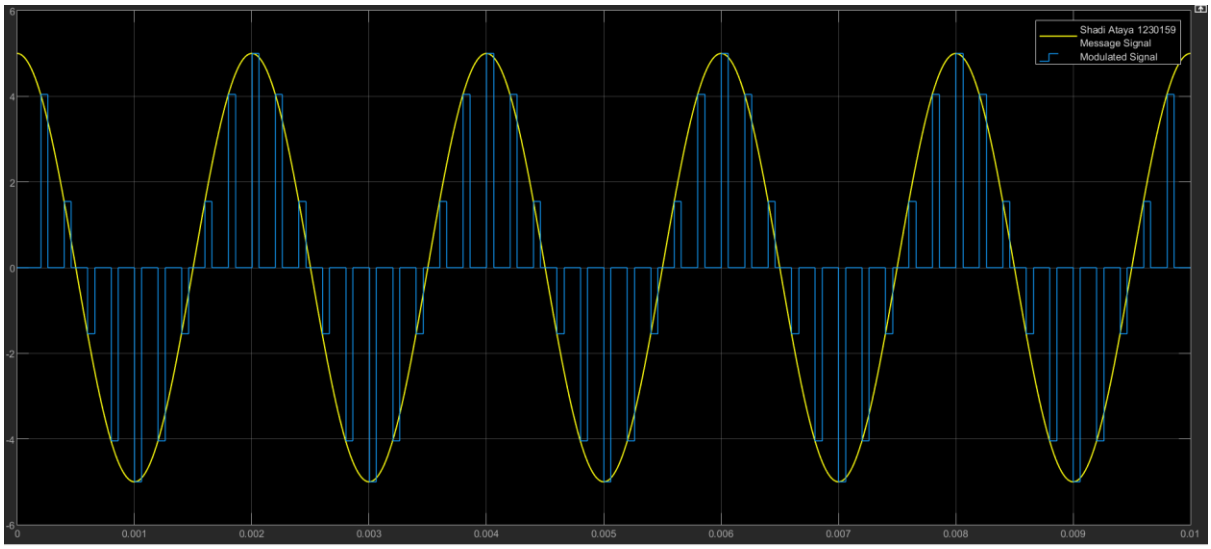


*Figure 36: Modulated Signal at frequency = **500Hz** and D = **30%** in time domain*

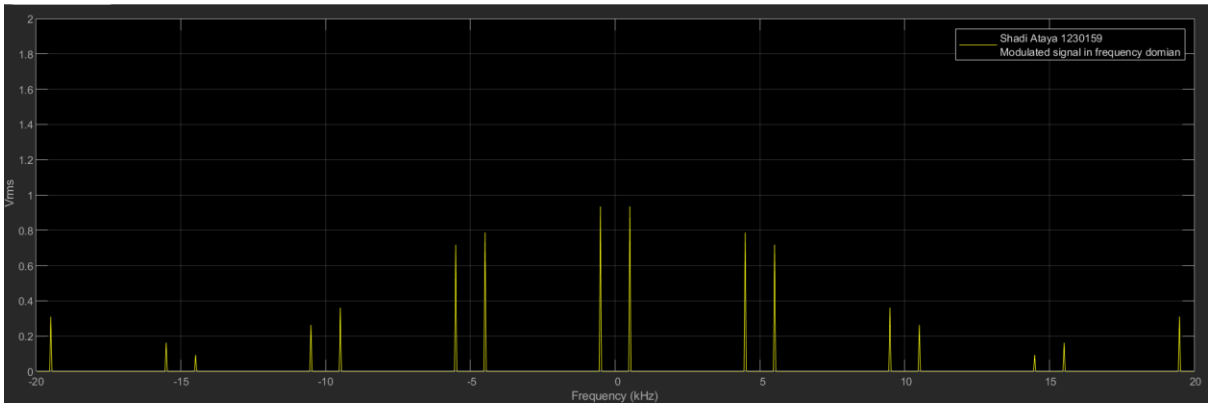


*Figure 37: Modulated Signal at frequency = **500Hz** and D = **30%** in frequency domain*

From Part 4: Modulated signal at frequency = 3000 and D = 50% in time domain and frequency domain

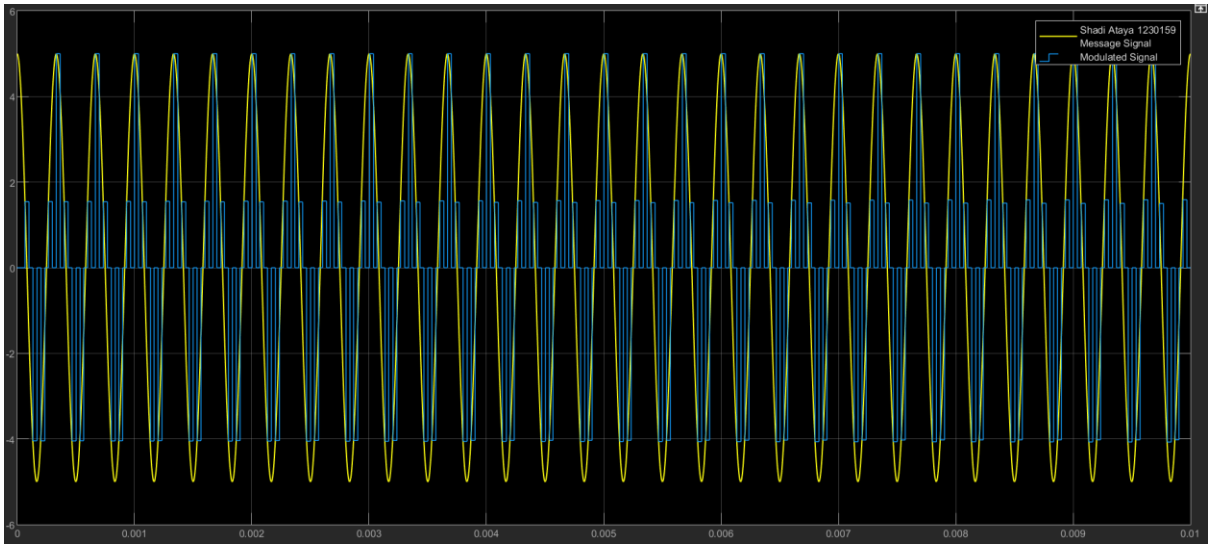


*Figure 38: Modulated signal Modulated signal at frequency = **3000** and D = **50%** in time domain*

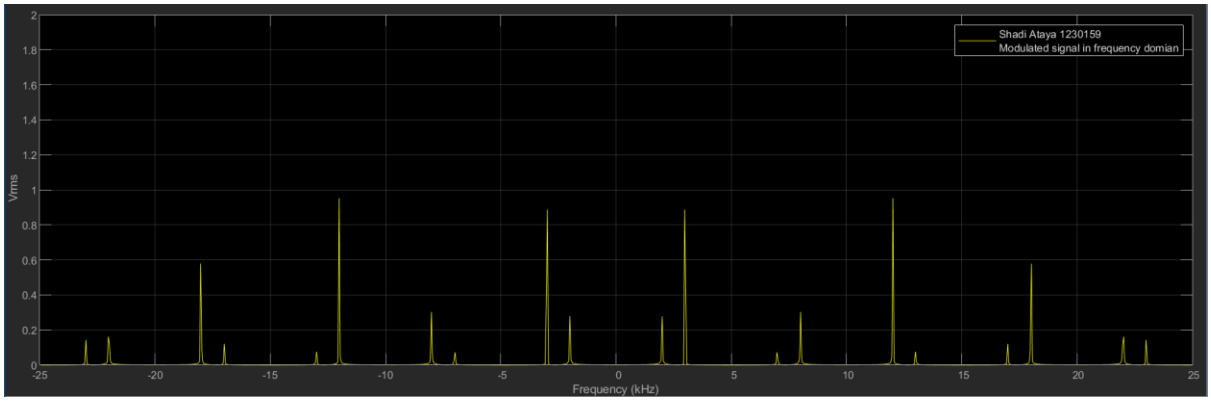


*Figure 39: Modulated signal Modulated signal at frequency = **3000** and D = **50%** in frequency domain*

Sample and Hold (Flat-top) Sampling Demodulation

Block Diagram:

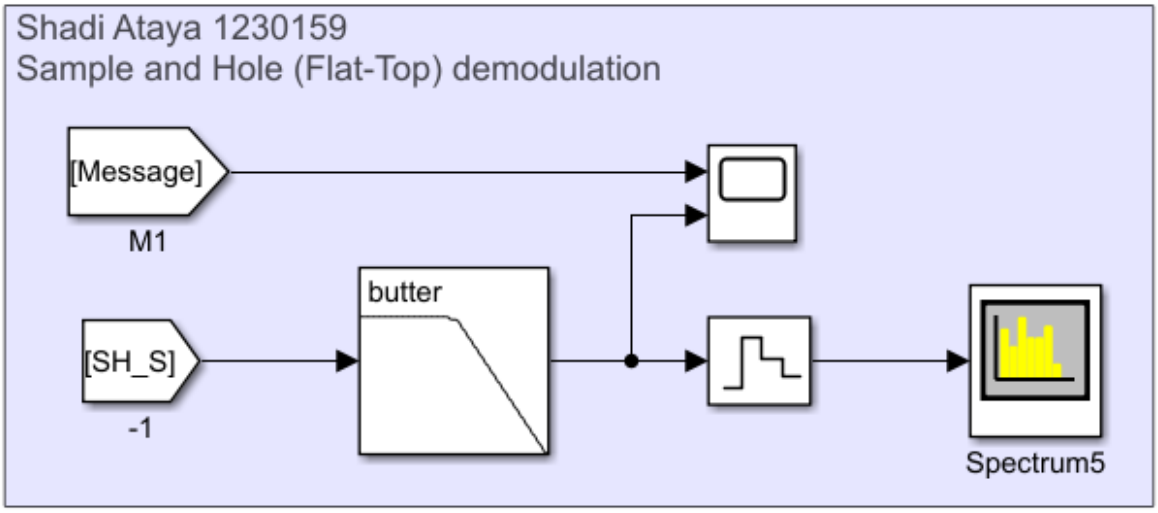


Figure 40: Sample and Hold (Flat-Top) Demodulation

Demodulated signal at frequency=500 and D =50% in time domain and frequency domain

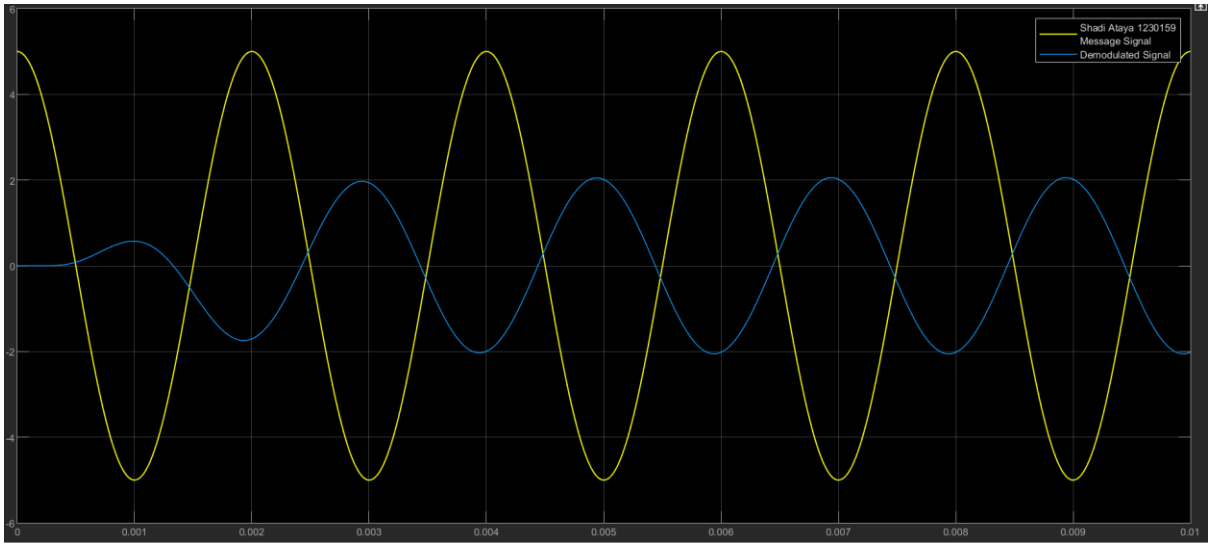


*Figure 41: Demodulated signal at frequency=**500** and D =**50%** in time domain*

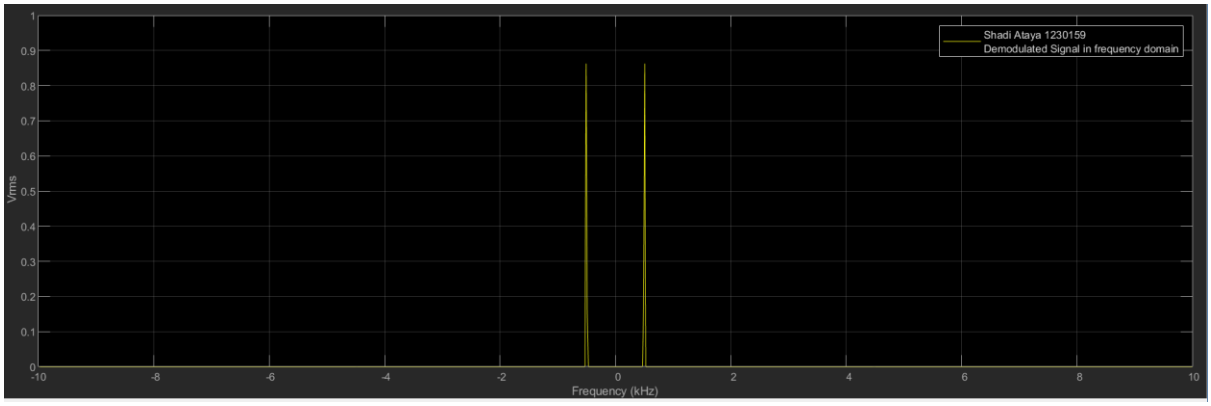


*Figure 42: Demodulated signal at frequency=**500** and D =**50%** in frequency domain*

From Part 4: Demodulated signal at frequency = 3000 and D = 50% in time domain and frequency domain

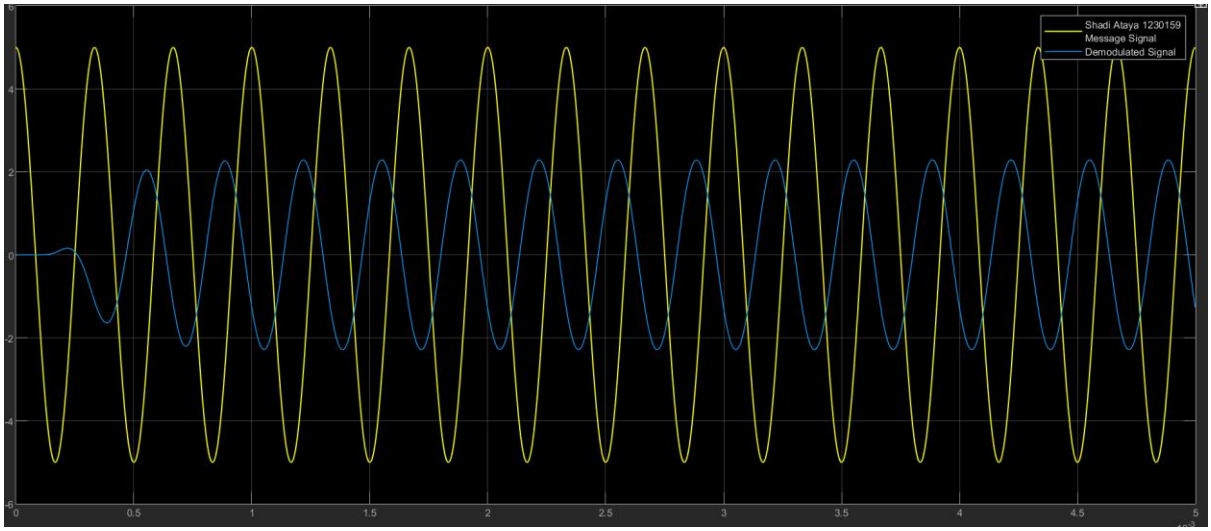


*Figure 43: Demodulated signal at frequency = **3000** and D = **50%** in time domain*

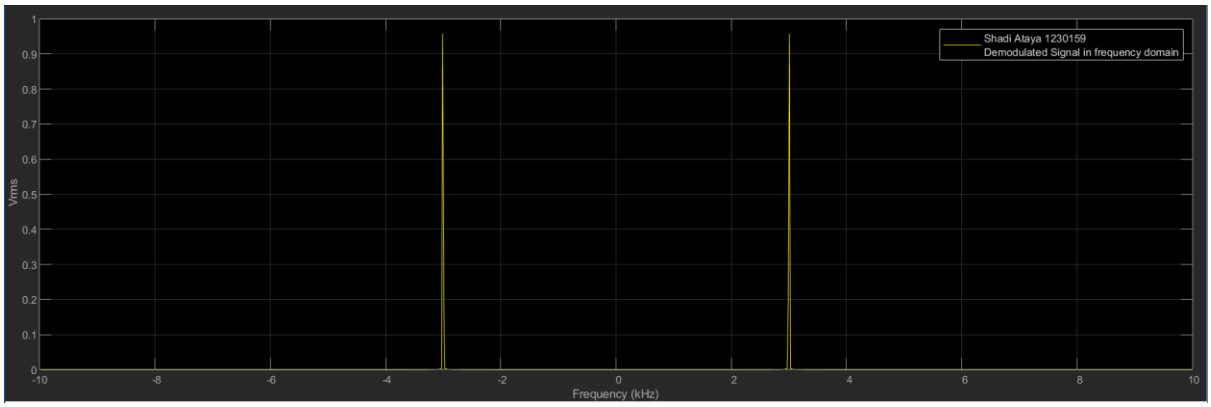


*Figure 44: Demodulated signal at frequency = **3000** and D = **50%** in frequency domain*